NEEDLE: A Live Benchmark of Web Search APIs for AI Agents

Ilya Gusev **Matthias Petri** **Andrey Styskin**
Keenable
{ilya.gusev,matthias,andrey}@keenable.ai

Abstract

NEEDLE is a live benchmark of web search APIs for AI agents. An agent sends such an API a text query and sees only the titles, URLs and snippets it returns. Agent benchmarks score the final answer, so they rarely test the search API alone. NEEDLE scores the API response directly. Hourly and daily runs build new queries from public sources for news, finance, research papers and law, and add a fixed stream of rare-word queries. We compare each engine with the best results that any tested engine found. Over eight weeks we test 26 search engines from 20 vendors. For 180 of 210 pairs of the 21 main engines, one engine scores reliably higher than the other on the same 12.4k queries. Searches by paper title hide large gaps: all but three engines find a paper from its title at recall@10 .71–.98, but from keywords in its body at only .08–.72. These results provide evidence that API comparisons should also test queries built from a paper's body.

1 Introduction

An **AI agent** is a language model that works in steps: it calls tools, reads their output and decides what to do next. Many agent tasks need facts from the web, such as a recent event, a company's quarterly income or a paper that another source cites. For these facts, the agent calls a **web search API**: a service that takes a text query and returns a short ranked list of results. Each result has a title, a URL and a snippet of page text. The agent sees only this list. It answers from the snippets or picks which pages to open. If the right page is not in the list, the agent cannot use it.

Many vendors sell such APIs, but a developer who must choose one has little public evidence. Agent benchmarks score the final answer, which mixes the search API with the model and the prompt (Section 2). Testing the API alone is hard.

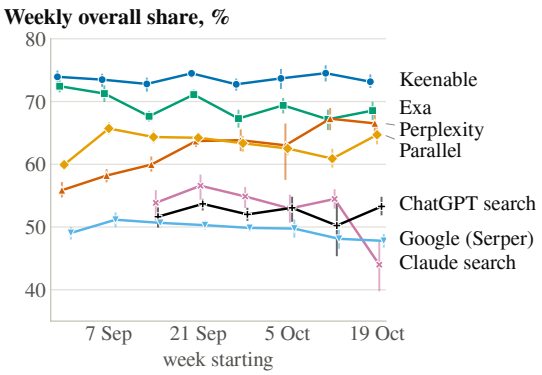


Figure 1: The top rank holds all eight weeks, but lower ranks change. Overall share by ISO week, 31 August–25 October 2026. ChatGPT and Claude search start on 14 September, and the pool grows from 18 to 25 engines. Points are query-weighted means, and bars are 95% run-level bootstrap intervals. Weekly scores count every failed news and rare request as zero (Appendix C).

The live web has no fixed document list or full relevance labels, and fixed test sets go stale and can leak into model training data (Sainz et al., 2023; Balloccu et al., 2024).

NEEDLE tests the search step alone. Each run sends the same new queries to every **engine**, which is one search API with fixed settings, and scores the results that each engine returns. The queries come from five tasks: news events, company finance facts, research papers, US law and rare-word queries from agent logs. For example, the scholar task looks for a paper with the query "gap rent regularity", whose words do not occur in the paper's title or abstract (Table 1). When a query has a known answer, such as a paper's arXiv id, we check whether the results contain it. Otherwise an LLM judge rates each result. We compare each engine with the **ceiling**: the best results that any tested engine found for that query.

We ran NEEDLE for eight weeks on 26 engines

Task	Query	Reference answer or profile
news	Paramount Warner Bros takeover closes 2026	judged: closing date, ephemeral event
finance, fact	nvidia number of employees	42,000, ±15%
finance, money	"servicenow" q3 fiscal 2024 operating income	$418M, ±0.1%
scholar, title	optimally selling heterogeneous objects by quantity	arXiv 2610.36244
scholar, body	gap rent regularity	same paper
legal, CFR	"modify or waive any" county committees authority	7 CFR 1417.2
rare	"Krasnogvardeysky District" Russia	judged

Table 1: Sample generated queries. News and rare have no reference answer. A judge rates their results against a query profile, shown here for news.

from 20 vendors. Our claims:

- **Most engine pairs have a clear winner.** On the same queries, one engine scores reliably higher than the other for most pairs of engines (Table 2).
- **Title searches hide large gaps.** Engines that find a paper from its title often miss it when the query is keywords from the paper's body (Figure 2).
- **Rankings are stable from week to week, but not fixed.** The same engine leads all eight weeks, but lower ranks change (Figure 1).

2 Related Work

Live and API benchmarks. LiveBench and LiveSearchBench draw new items from dated sources but score model answers (White et al., 2025; Zhou et al., 2026a). NEEDLE scores the search results behind such answers. BrowseComp scores agents end to end (Wei et al., 2025). FreshStack and BrowseComp-Plus fix the corpus to isolate retrieval (Thakur et al., 2025; Chen et al., 2026), while NEEDLE keeps the live web. API comparisons use static questions under one agent (Selvam and Ghosh, 2026; Artificial Analysis, 2026), and web engine audits use fixed query sets (Lewandowski, 2015; Kirsten et al., 2026). NEEDLE draws new queries each run and scores the API without an agent.

Scorers and tasks. LLM assessors and simulated known-item queries can reproduce system rankings (Thomas et al., 2024; Upadhyay et al., 2025; Azzopardi et al., 2007; He et al., 2025). NEEDLE uses both and checks its judge against models from other vendors. Domain benchmarks cover literature, caselaw and finance search on fixed sets (Ajith et al., 2024; Hou et al., 2025; Hu et al., 2026), while NEEDLE runs these tasks on new queries each day. Full text helps retrieval beyond abstracts (Lin, 2009), and our body queries test that gap on live APIs.

3 The Benchmark

We test 26 engines from 20 vendors. The 21 **main engines** are one engine per vendor plus Brave's LLM-context endpoint. The other five are mode variants, such as Exa Instant.

A **query-run** is one query issued in one run. We use three windows in 2026: eight weeks, 31 August–25 October (Figure 1), the **main window**, 17–28 October, with the 276 runs in which all main engines took part (Table 2), and 31 scholar runs, 29 September–28 October (Figure 2). Table 3 lists the tasks, and Appendix A the API settings and data sources.

3.1 Queries

News. Each hour an LLM writes 20 event queries from news feeds and Google Trends, rejecting evergreen items.

Finance. **Fact** queries ask for company facts from Wikidata (Vrandečić and Krötzsch, 2014) and GLEIF, **money** queries for quarterly 10-Q figures from SEC XBRL, and **filing** queries for one SEC filing by description.

Scholar. Each recent arXiv or Europe PMC (Europe PMC Consortium, 2015) paper yields four paired known-item queries (Azzopardi et al., 2007): its normalized **title**, **body keywords** for a detail absent from title and abstract, a **body question** about that detail, and a **tip-of-tongue** description (Arguello et al., 2021) without names or exact values. All four must pass the filters.

Legal. CourtListener opinions yield caption queries, and eCFR sections yield queries based on verbatim spans. Filters reject citation leaks.

Rare. We sample a fixed public stream of English agent queries containing rare words (Ning et al., 2026; Li et al., 2026; Zhou et al., 2026b). Table 1 gives query examples, and Appendix B gives the filters.

At least half of the legal queries and of the fi-

Engine	Overall share, %		Scholar	News	Finance	Rare	Legal	Failed, %	Latency
	mean	95% CI	R@10	nDCG@5	R@5	nDCG@5	R@5	requests	p_{50}, ms
ceiling (pooled)	*100*		*.901*	*1*	*.993*	*1*	*.971*		
Keenable	73.2	72.3–74.1	**.731**	.609	.903	**.508**	**.797**	0.0	135
Exa	68.6	67.5–69.5	.563	.604	**.922**	.476	.771	0.3	2,132
Perplexity	66.9	65.3–68.5	.578	.640	.895	.468	.674	0.0	433
Parallel	65.1	63.9–66.3	.587	.546	.907	.387	.734	0.8	2,660
Brave (LLM ctx)	64.6	63.4–65.6	.533	**.642**	.858	.454	.658	0.0	1,112
Linkup	57.6	56.1–58.9	.470	.567	.797	.400	.570	0.0	1,212
Brave	57.3	56.2–58.4	.488	.557	.769	.354	.621	0.0	487
Context	57.1	55.4–58.6	.598	.469	.782	.339	.576	0.0	866
DeepSeek	56.9	55.8–57.9	.271	.635	.818	.387	.676	0.0	551
You.com	55.8	54.7–56.9	.471	.553	.747	.343	.602	0.0	486
ChatGPT search†§	52.4	51.0–53.8	.300	.354	.809	.400	.697	11.1	43,831
Kagi§	52.1	51.1–53.2	.560	.318	.728	.299	.617	7.6	1,275
Claude search†§	51.3	49.6–53.0	.498	.462	.685	.322	.525	35.9	33,906
Firecrawl	50.7	49.2–52.0	.488	.286	.820	.265	.598	0.0	688
Octen	50.2	48.8–51.7	.307	.511	.730	.322	.586	0.0	197
Google (Serper)	47.7	46.8–48.6	.475	.262	.738	.255	.582	0.0	1,785
TinyFish§	45.1	43.6–46.4	.429	.256	.730	.248	.521	29.0	1,904
Bing (SearchAPI)	44.1	41.1–46.8	.416	.288	.675	.173	.586	0.4	2,008
Jina	43.8	42.4–45.1	.492	.210	.726	.163	.523	4.9	17,272
Tavily	42.6	41.5–43.7	.303	.303	.828	.284	.361	0.1	1,999
Ceramic	14.7	13.9–15.7	.001	.040	.441	.109	.137	0.1	99

Table 2: Results on 276 shared runs, 17–28 October 2026 (12.4k query-runs, 21 engines). The ceiling also includes variants. Task cells are absolute metrics, with the best in bold. Italics mark the ceiling. Scores exclude account-side failures, and Failed counts all request failures. Latency is the per-run median on news. †Server-side search tool, so latency includes answer generation. §Account-side exclusions leave rank uncertain (Appendix C).

nance money and filing queries carry a search operator: quotes, `site:`, `after:` or `before:`. Where an API has a native filter for an operator, the client uses it.

Task	Cadence	Queries/run	k	Metric
news	hourly	20	5	nDCG@5
finance	daily	120	5	recall@5
scholar	daily	up to 420	10	recall@10
legal	daily	40 + 0–20	5	recall@5
rare	daily	100	5	nDCG@5

Table 3: Benchmark tasks, run frequency and primary metrics. k is the number of search results requested per query. Query counts can fall below the targets after source and query filtering. Legal combines 40 CFR queries with caption queries for recent court opinions.

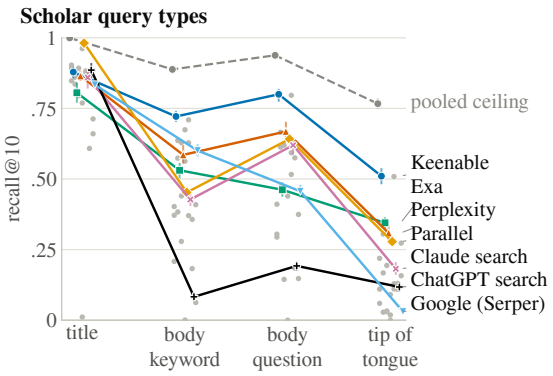


Figure 2: Engines that find papers from titles differ widely on body queries. Scholar recall@10 by query type: 31 runs, 29 September–28 October 2026, with 2,600–3,005 query-runs per line and type. Lines show seven main engines, grey dots the other 19 engines, and dashes the pooled ceiling. Points are query-weighted means, and bars are 95% run-level bootstrap intervals.

3.2 Scores

News and rare. GPT-6 Luna at high reasoning effort rates each distinct result from 0 to 4 using a shared query profile, with checks for relevance, coverage, freshness and literal operator compliance. The scale adapts the Needs Met levels of Google (2025). Ratings 0–4 map to gains $g = \{0, 0, .117, .667, 1\}$, which put high weight on strong results (Kanoulas and Aslam, 2009): one rating-4 result is worth more than eight rating-2 results. The judge sees at most 2,000 characters of each snippet.

For each query, the ceiling keeps the best rating per normalized URL across all engines, variants included, and sorts these ratings (Spärck Jones and van Rijsbergen, 1975; Zobel, 1998). With r_i the rating at rank i, we compute nDCG@5 (Järvelin and Kekäläinen, 2002) as

$$\mathrm{nDCG}_e(q) = \frac{1}{\mathrm{IDCG}(q)} \sum_{i=1}^{5} \frac{g(r_i)}{\log_2(i+1)}.$$

IDCG is the same sum over the top five pooled rat-

ings. Missing ranks contribute zero. Repeated normalized URLs receive zero gain except on `site:` queries. We drop a query-run from all engines if any judgment fails or no result reaches rating 2. The task score is the mean over retained query-runs.

Scholar, legal and finance. Recall@k counts queries with a matching result in the top k. Scholar matches arXiv ids, DOIs and PMIDs. Legal matches citations, dockets and identifiers in URLs and snippets. Caselaw recall measures case identity, not necessarily the opinion page. Finance checks whether the snippets contain the answer, with a 0.1% money band and a 15% headcount band. The ceiling is a hit by any engine. Appendix F gives the matching rules and checks matcher error.

3.3 Overall share and failures

For task metric M_t, an engine's overall share is

$$S(e) = \frac{1}{5}\sum_{t=1}^{5}\frac{M_t(e)}{M_t(\text{ceiling})}.$$

The tasks have equal weight. The ceiling is the best the tested engines found, not the best the web contains.

We exclude account-side failures (our quota, rate limit, login or credit) on every task, and other failures score zero. The exclusions remove up to 36% of an engine's query-runs (Appendix C).

Comparing two engines. Some queries are hard for every engine, so intervals for single engines can overlap even when one engine beats another on the same queries. We therefore bootstrap the difference $S(a) - S(b)$. Within each task, we resample whole runs 2,000 times, using the same runs for every engine (Efron and Tibshirani, 1993), and recompute the overall shares. Each difference gets a normal-approximation interval with Bonferroni correction for all $\binom{21}{2} = 210$ pairs at a 5% familywise level (Carterette, 2012). An interval that excludes zero orders that pair. The intervals measure sampling uncertainty, not scorer error (Appendix C).

3.4 Judge validation

We sample 100 news and 100 rare queries, with six distinct top-five URLs per query (1,200 results). Four models judge them with the production prompt, input and settings: GPT-6 Luna (the production judge), Claude Opus 5.5, Gemini 4 Argon and Grok 4.7. We score each model against the mean grade of the other three. GPT-6 Luna agrees with this leave-one-out consensus at quadratic-weighted $\kappa = .89$ (Cohen, 1968), and the other three models at .91–.92 (Table 7, Appendix E). Its mean grade of 1.59 is inside the 1.55–1.67 range of the others.

In a second check, Gemini and Grok grade the complete result pools of 100 queries (5,698 results from all 21 main engines) from snippets, with a separate written guideline. The engine ranking from each rater agrees with the ranking from the production judge at Spearman .94. Neither rater separates Keenable from Exa on this sample. Both checks measure model agreement, not human validation.

4 Results

Most engine pairs have a clear winner, but not all. For 180 of the 210 pairs of main engines, the adjusted interval for the score difference excludes zero, so one engine scores reliably higher than the other. Of these, 169 also hold when we resample days instead of runs and when we keep only query-runs with no account-side failure in any main engine (Appendix C). The other 30 pairs stay unordered, which does not mean equal scores. No engine leads every task (Table 2), so the overall lead depends on the task mix. Keenable leads Exa by 4.6 points, and scholar contributes 3.7 of them. With scholar dropped and the other four tasks at equal weight, the lead is 1.1 points.

Title searches hide large gaps. An agent that follows up a detail it read elsewhere has no title to search for. Over 31 scholar runs, all engines but Ceramic, DeepSeek and Octen reach recall@10 .71–.98 from titles. The same 23 engines span .08–.72 on body keywords (Figure 2 and Table 6), so title tests can overstate how well an engine serves such an agent. Keenable leads at .72, followed by its realtime mode (.71) and Kagi (.67). Appendix D compares engines on common papers.

Rankings are stable from week to week, but not fixed. Adjacent weekly rankings agree at Kendall's τ_b (Kendall, 1945) .81–.94. Keenable leads all eight weeks, but lower ranks change (Figure 1, checks in Appendix C).

Limitations

Conflict of interest. The authors' organization builds an evaluated engine. The code, prompts,

clients and per-query outputs are public so readers can audit the claims. All engines receive the same queries, snippet cap, request pacing and judge. Each keeps one setting throughout the main window. We cannot tell whether a vendor changed its index for the benchmark or for other reasons.

Evidence and judge. Scores measure the API response, not the fetched page. Short snippets can hide relevant content (Turpin et al., 2009; Kazai et al., 2022). News and rare depend on one LLM judge. All audit raters are also LLMs, which cannot replace human validation (Faggioli et al., 2023; Clarke and Dietz, 2025). LLM judges can reward query-term stuffing (Alaofi et al., 2024) and favor some vendors or their own outputs (Balog et al., 2025; Soboroff, 2025; Panickssery et al., 2024). The judge shares a vendor with ChatGPT search, and some audit raters share one with Claude search.

Matcher error. Identifier matching can miss a target page that lacks its id. Finance table figures can match the wrong period when snippets omit column headers. Appendix F bounds this error, but only pooled over engines.

Coverage and weights. Queries are English and US-centric. Equal task weights do not estimate agent traffic, and the lead depends strongly on scholar. Default API tiers need not be each vendor's best. ChatGPT and Claude search return sources through answer-generating tools, so their quality scores are comparable but their latency and cost are not. The ceiling depends on the engine pool (Buckley et al., 2007), so compare shares only within a common window.

Failures and timing. Excluding account-side failures assumes those requests would score like the rest. Without that assumption, Claude search can take any rank from 3 to 20 and TinyFish from 11 to 20 (Appendix C). Engines process queries at different speeds, so slow engines see news later. A five-minute dispatch filter leaves the news order unchanged.

Reproducibility. Public sources and queries permit tuning or special handling of benchmark traffic. The archive holds the queries, reference answers, results and judgments of the studied runs, but not each run's code revision. Appendix A gives the source terms.

References

Anirudh Ajith, Mengzhou Xia, Alexis Chevalier, Tanya Goyal, Danqi Chen, and Tianyu Gao. 2024. LitSearch: A retrieval benchmark for scientific literature search. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 15068–15083.

Marwah Alaofi, Paul Thomas, Falk Scholer, and Mark Sanderson. 2024. LLMs can be fooled into labelling a document as relevant: best café near me; this paper is perfectly relevant. In *Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP)*, pages 32–41.

Jaime Arguello, Adam Ferguson, Emery Fine, Bhaskar Mitra, Hamed Zamani, and Fernando Diaz. 2021. Tip of the tongue known-item retrieval: A case study in movie identification. In *Proceedings of the 2021 Conference on Human Information Interaction and Retrieval (CHIIR)*, pages 5–14.

Artificial Analysis. 2026. Artificial analysis search index: Same agent, different search. Launched August 18, 2026; accessed November 3, 2026.

Leif Azzopardi, Maarten de Rijke, and Krisztian Balog. 2007. Building simulated queries for known-item topics: An analysis using six European languages. In *Proceedings of the 30th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 455–462.

Simone Balloccu, Patrícia Schmidtová, Mateusz Lango, and Ondřej Dušek. 2024. Leak, cheat, repeat: Data contamination and evaluation malpractices in closed-source LLMs. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 67–93.

Krisztian Balog, Donald Metzler, and Zhen Qin. 2025. Rankers, judges, and assistants: Towards understanding the interplay of LLMs in information retrieval evaluation. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 3865–3875.

Chris Buckley, Darrin Dimmick, Ian Soboroff, and Ellen M. Voorhees. 2007. Bias and the limits of pooling for large collections. *Information Retrieval*, 10(6):491–508.

Benjamin A. Carterette. 2012. Multiple testing in statistical analysis of systems-based information retrieval experiments. *ACM Transactions on Information Systems*, 30(1):4:1–4:34.

Zijian Chen, Xueguang Ma, Shengyao Zhuang, Ping Nie, Kai Zou, Sahel Sharifymoghaddam, et al. 2026. BrowseComp-Plus: A fair and disentangled evaluation benchmark for deep search agents. In *Proceedings of the 64th Annual Meeting of the Association*

for Computational Linguistics (Volume 1: Long Papers), pages 22349–22370.

Charles L. A. Clarke and Laura Dietz. 2025. LLM-based relevance assessment still can't replace human relevance assessment. In *Proceedings of the International Workshop on Evaluating Information Access (EVIA 2025), a Satellite Workshop of the NTCIR-18 Conference*, pages 1–5.

João Coelho, Jingjie Ning, Jingyuan He, Kangrui Mao, Abhijay Sai Paladugu, Pranav Setlur, et al. 2026. DeepResearchGym: A free, transparent, and reproducible sandbox for deep research. In *Proceedings of the 2026 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR)*, pages 34–43.

Jacob Cohen. 1968. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4):213–220.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186.

Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*, volume 57 of *Monographs on Statistics and Applied Probability*. Chapman & Hall, New York.

Europe PMC Consortium. 2015. Europe PMC: a full-text literature database for the life sciences and platform for innovation. *Nucleic Acids Research*, 43(D1):D1042–D1048.

Guglielmo Faggioli, Laura Dietz, Charles L. A. Clarke, Gianluca Demartini, Matthias Hagen, Claudia Hauff, et al. 2023. Perspectives on large language models for relevance judgment. In *Proceedings of the 2023 ACM SIGIR International Conference on the Theory of Information Retrieval (ICTIR)*, pages 39–50.

Google. 2025. *Search Quality Rater Guidelines: General Guidelines*. Version of September 11, 2025.

Yifan He, To Eun Kim, Fernando Diaz, Jaime Arguello, and Bhaskar Mitra. 2025. Tip of the tongue query elicitation for simulated evaluation. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 3398–3407.

Abe Bohan Hou, Orion Weller, Guanghui Qin, Eugene Yang, Dawn Lawrie, Nils Holzenberger, et al. 2025. CLERC: A dataset for U.S. legal case retrieval and retrieval-augmented analysis generation. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 7913–7928.

Liang Hu, Jianpeng Jiao, Jiashuo Liu, Dongyuan Mutu, Yanle Ren, Zhoufutu Wen, et al. 2026. FinSearchComp: Towards a realistic, expert-level evaluation of financial search and reasoning. In *The Fourteenth International Conference on Learning Representations (ICLR)*.

Kalervo Järvelin and Jaana Kekäläinen. 2002. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446.

Armand Joulin, Edouard Grave, Piotr Bojanowski, Matthijs Douze, Hervé Jégou, and Tomas Mikolov. 2016. FastText.zip: Compressing text classification models. *Preprint*, arXiv:1612.03651.

Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. Bag of tricks for efficient text classification. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 427–431.

Evangelos Kanoulas and Javed A. Aslam. 2009. Empirical justification of the gain and discount function for nDCG. In *Proceedings of the 18th ACM Conference on Information and Knowledge Management (CIKM)*, pages 611–620.

Gabriella Kazai, Bhaskar Mitra, Anlei Dong, Nick Craswell, and Linjun Yang. 2022. Less is less: When are snippets insufficient for human vs machine relevance estimation? In *Advances in Information Retrieval: 44th European Conference on IR Research (ECIR 2022), Part II*, pages 153–162.

M. G. Kendall. 1945. The treatment of ties in ranking problems. *Biometrika*, 33(3):239–251.

Elisabeth Kirsten, Jost Grosse Perdekamp, Qinyuan Wu, Mihir Upadhyay, Krishna P. Gummadi, and Muhammad Bilal Zafar. 2026. Characterizing web search in the age of generative AI. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 10827–10848.

Dirk Lewandowski. 2015. Evaluating the retrieval effectiveness of web search engines using a representative query sample. *Journal of the Association for Information Science and Technology*, 66(9):1763–1775.

Zhuofeng Li, Dongfu Jiang, Xueguang Ma, Haoxiang Zhang, Ping Nie, Yuyu Zhang, Kai Zou, Jianwen Xie, Yu Zhang, and Wenhu Chen. 2026. OpenResearcher: A fully open pipeline for long-horizon deep research trajectory synthesis. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, pages 908–937.

Jimmy Lin. 2009. Is searching full text more effective than searching abstracts? *BMC Bioinformatics*, 10:46.

Jingjie Ning, João Coelho, Yibo Kong, Yunfan Long, Bruno Martins, João Magalhães, Jamie Callan, and Chenyan Xiong. 2026. Agentic search in the wild: Intents and trajectory dynamics from 14m+ real search requests. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1404–1415.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, pages 68772–68802.

Oscar Sainz, Jon Ander Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 10776–10787.

Sriram Selvam and Anneswa Ghosh. 2026. Similar accuracy, unequal evidence: Search APIs as decision surfaces for tool-using agents. In *Proceedings of the 1st Workshop on Grounding Language Models: Learning Faithfully and Efficiently (GroundLM 2026)*, pages 606–622.

Ian Soboroff. 2025. Don't use LLMs to make relevance judgments. *Information Retrieval Research*, 1(1):29–46.

Karen Spärck Jones and C. J. van Rijsbergen. 1975. Report on the need for and provision of an 'ideal' information retrieval test collection. British Library Research and Development Report 5266, Computer Laboratory, University of Cambridge.

Nandan Thakur, Jimmy Lin, Sam Havens, Michael Carbin, Omar Khattab, and Andrew Drozdov. 2025. FreshStack: Building realistic benchmarks for evaluating retrieval on technical documents. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

Paul Thomas, Seth Spielman, Nick Craswell, and Bhaskar Mitra. 2024. Large language models can accurately predict searcher preferences. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1930–1940.

Andrew Turpin, Falk Scholer, Kalervo Järvelin, Mingfang Wu, and J. Shane Culpepper. 2009. Including summaries in system evaluation. In *Proceedings of the 32nd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 508–515.

Shivani Upadhyay, Ronak Pradeep, Nandan Thakur, Daniel Campos, Nick Craswell, Ian Soboroff, and Jimmy Lin. 2025. A large-scale study of relevance assessments with large language models using UMBRELA. In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR)*, pages 358–368.

Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10):78–85.

Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, et al. 2025. BrowseComp: A simple yet challenging benchmark for browsing agents. *Preprint*, arXiv:2504.12516.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, et al. 2025. LiveBench: A challenging, contamination-limited LLM benchmark. In *International Conference on Learning Representations (ICLR)*.

Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi, Wolfgang Macherey, et al. 2016. Google's neural machine translation system: Bridging the gap between human and machine translation. *Preprint*, arXiv:1609.08144.

Heng Zhou, Ao Yu, Yuchen Fan, Li Kang, Jianing Shi, Yongting Zhang, et al. 2026a. An automated pipeline for provably retrieval-dependent benchmark construction over dynamic knowledge. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, pages 40633–40648.

Yuqi Zhou, Sunhao Dai, Changle Qu, Liang Pang, Jun Xu, and Ji-Rong Wen. 2026b. Learning to retrieve from agent trajectories. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2544–2555.

Justin Zobel. 1998. How reliable are the results of large-scale information retrieval experiments? In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 307–314.

A API Settings and Data Sources

All engines receive the same query and requested result count. Each engine processes one request at a time, and engines run concurrently. Requests have a 30-second timeout and up to three attempts, with exponential back-off from 1 to 30 seconds on HTTP 429, 5xx and timeouts. Short result lists score as returned.

The generators and judge use GPT-6 Luna (`openai/gpt-6-luna`) through OpenRouter, with temperature 0 and high reasoning effort. Table 4 lists the search clients and settings. Appendix G links the prompts.

The rare stream draws from DeepResearchGym (Coelho et al., 2026; Ning et al., 2026) (59%, CC BY 4.0), OpenResearcher (Li et al., 2026) (39%,

Vendor	Endpoint	Variant or non-default parameters
Keenable	api.keenable.ai	default, realtime variant
Exa	api.exa.ai	`type=auto`, instant and lite variants
Google	google.serper.dev (Serper)	default
Bing	searchapi.io (SearchAPI)	engine `bing`
Brave	api.search.brave.com	web search, LLM-context endpoint
Parallel	api.parallel.ai	default processor, turbo variant
Perplexity	api.perplexity.ai	search API, fast variant
Tavily	api.tavily.com	default
Linkup	api.linkup.so	default
TinyFish	api.search.tinyfish.ai	default
Octen	api.octen.ai	default
Ceramic	api.ceramic.ai	default
You.com	ydc-index.io	default
Firecrawl	api.firecrawl.dev	search endpoint, default
Kagi	kagi.com/api/v1	default
Jina	s.jina.ai	default
Context	api.context.dev/v1	default
DeepSeek	api.deepseek.com	`/v1/tools/search`, default
Claude search	api.anthropic.com	`claude-sonnet-5`, `web_search`
ChatGPT search	api.openai.com	`gpt-5.5`, `web_search`, `search_context_size=low`

Table 4: Search clients and settings by vendor. Clients live in `src/needle/shared/search/` and also implement the native query filters.

MIT) and LRAT-Train (Zhou et al., 2026b) (1%, no stated license). Source terms govern query texts. Our code is MIT, and our CC BY 4.0 license covers results, judgments and metadata, not source text. LRAT contributes 17 of 1,200 main-window rare queries, and removing them changes no mean rare nDCG@5 by more than .003. Queries can contain personal information, and we accept private removal requests. API output records remain subject to vendor terms.

B Query Filters

Scholar samples five domains (computer, physical, life, social and health sciences) and three age bands (7 days, 30 days, 1 year), targeting seven accepted papers per cell. Each paper must pass all four filters:

- **Title:** at least four words, plus a digit, acronym, hyphenated term or five content tokens.
- **Body keywords:** at least three words and two content tokens absent from title and abstract.
- **Body question:** at least eight words and the same novelty test as body keywords.
- **Tip-of-tongue:** at least twelve words and at most two content tokens shared with the title, with no quotes, decimal, five-digit number or coined title/abstract term.

One logged run accepts 323 of 572 candidate papers (56%). The effect of rejecting other papers on engine order is untested.

News draws from 124 feeds and Google Trends RSS for 52 US geographies. Prompts reject evergreen items and request two-to-six-word event queries. Legal captions come from 14 federal courts on CourtListener. CFR queries contain one verbatim two-to-eight-word span but no citation or target section number. Only the quoted variant retains quotation marks. Rare queries must pass fastText English language ID (Joulin et al., 2017, 2016) and contain a word split into at least five BERT WordPieces (Wu et al., 2016; Devlin et al., 2019).

C Statistical Checks and Ranking Sensitivity

Paired intervals. The main window has 12 scholar, 232 news, 10 finance, 12 rare and 10 legal runs. For pair (a, b), let $\widehat{\Delta}_{ab} = S(a) - S(b)$ and let s_{ab} be the standard deviation of its 2,000 bootstrap differences. The adjusted interval is $\widehat{\Delta}_{ab} \pm z\, s_{ab}$ with $z = 3.68$, the two-sided Bonferroni value for 210 comparisons at familywise level .05. The single-engine intervals in Table 2 instead use the 2.5th and 97.5th bootstrap percentiles. Their overlap is not a test of the difference between two engines.

Dependence and missing requests. We repeat the comparisons by resampling days, and by clustering exact query text (target paper for scholar) across runs. The latter checks repeated targets, which day resampling alone does not address. We

Engine	All	Matched	Δ matched	95% CI for Δ
Keenable	73.2	74.7	+1.6	+0.4, +2.7
Exa	68.6	67.5	−1.1	−2.3, +0.2
Perplexity	66.9	66.2	−0.7	−2.5, +1.1
Parallel	65.1	63.0	−2.1	−3.5, −0.7
Brave (LLM ctx)	64.6	67.1	+2.5	+1.1, +3.9
Linkup	57.6	59.3	+1.7	−0.1, +3.8
Brave	57.3	59.1	+1.8	+0.2, +3.2
Context	57.1	56.2	−0.9	−3.0, +1.1
DeepSeek	56.8	58.0	+1.1	−0.6, +2.9
You.com	55.8	57.4	+1.6	+0.3, +2.9
ChatGPT search	52.4	51.9	−0.5	−2.0, +0.8
Kagi	52.1	52.3	+0.1	−0.8, +1.2
Claude search	51.4	52.6	+1.2	−0.4, +2.9
Firecrawl	50.7	50.1	−0.5	−1.9, +0.9
Octen	50.2	49.1	−1.1	−2.7, +0.6
Google (Serper)	47.7	47.4	−0.3	−1.5, +0.9
TinyFish	45.1	44.7	−0.4	−2.0, +1.2
Bing (SearchAPI)	44.1	48.3	+4.1	+1.5, +7.5
Jina	43.8	43.2	−0.5	−2.2, +1.2
Tavily	42.6	42.5	−0.1	−1.2, +1.3
Ceramic	14.7	14.7	−0.1	−1.1, +1.2

Table 5: Overall shares in percent. All uses the 12,313 query-runs of the 274 main-window runs that have per-query files, so it can differ from Table 2 by 0.1. Matched restricts these to 4,048 query-runs with no account failure in any main engine. Δ is Matched minus All, in percentage points, with a paired run-bootstrap interval.

also restrict the data to query-runs with no account-side failure in any main engine. Of the 180 pairs whose primary adjusted intervals exclude zero, 169 also exclude zero under both day resampling and this matched-support check.

Table 5 shows the effect of matching, which retains 4,048 of the 12,313 query-runs that have per-query files. Claude search moves from 51.4 to 52.6. The matched order has $\tau_b = .91$ against Table 2, with Keenable and Exa still first and second. If one engine's excluded requests instead score zero or at the ceiling, Claude search can take any rank from 3 to 20, TinyFish from 11 to 20, Kagi from 10 to 16 and ChatGPT search from 9 to 15. Missing requests therefore limit the rank claims for these four engines.

The published per-run scores, which Figure 1 uses, count account-side failures on news and rare as zero. Under that rule, Claude search scores 45.1 instead of 51.3, and no other engine moves by more than 0.9 points.

Normalization, weights and timing. Removing an engine's own results from the ceiling changes its overall share by at most 0.8 points. Dropping any one task preserves the leader. The Keenable–Exa gap then ranges from 1.1 points (without scholar) to 6.2 points (without finance).

Among the 15 engines present in all eight plotted weeks, adjacent rankings have $\tau_b = .85$–$.94$. Fixing the news pool to those 15 changes no sampled weekly order. Restricting news comparisons to requests sent within five minutes of the first engine retains 96% of scored pairs, leaves the order unchanged and moves no mean nDCG@5 by more than .010. These checks address changing engine participation and dispatch lag.

D Scholar Query Types and Paired Differences

Table 6 gives every engine and its sample size. In the main window, Keenable's recall lead over Exa is .38 for 7-day papers, .10 for 30-day papers and .03 for one-year papers. In health sciences, the lead is negative (−.10). The advantage is not uniform across age or domain.

For paired query-type comparisons, we retain 2,629 paper-runs from 29 runs where Keenable, its realtime mode, Kagi, Exa and ChatGPT search all have scores for all four types (10,516 query-runs). Account-side failures exclude the paper from all five engines, and other failures score zero. We use stored identifier hits and 2,000 paired run-bootstrap resamples.

Body-keyword recall is .154 lower than title recall for Keenable and .802 lower for ChatGPT search. On body keywords, Keenable leads its realtime mode by .012, Kagi by .048, Exa by .199 and ChatGPT search by .644. The 95% intervals of

Engine	Title	Body keyword	Body question	Tip of tongue	Runs	Query-runs
ceiling (pooled)	*1*	*.888*	*.938*	*.767*	*31*	*12,020*
Parallel	**.981**	.453	.642	.278	31	12,020
Kagi	.962	.674	.640	.091	30	11,599
Bing (SearchAPI)	.903	.402	.411	.060	31	12,020
Parallel Turbo	.894	.208	.294	.110	31	12,020
Brave (LLM ctx)	.893	.440	.620	.194	31	12,020
ChatGPT search	.886	.083	.192	.118	30	11,016
You.com	.883	.370	.593	.129	31	12,020
Brave	.881	.383	.613	.167	31	12,020
Keenable	.879	**.721**	**.800**	**.510**	31	12,020
Linkup	.878	.357	.575	.130	21	7,898
Keenable realtime	.877	.709	.796	.509	31	12,020
Perplexity	.865	.585	.668	.309	30	11,636
Jina	.862	.629	.470	.030	31	11,716
Perplexity Fast	.860	.574	.666	.305	30	11,636
Claude search	.860	.427	.620	.182	27	10,400
Context	.856	.628	.516	.111	31	12,020
Exa Lite	.848	.371	.305	.219	1	420
Firecrawl	.836	.636	.472	.028	31	12,020
Google (Serper)	.834	.601	.457	.030	31	12,020
Exa	.805	.531	.462	.345	31	12,020
Tavily	.795	.279	.144	.033	31	12,020
TinyFish	.769	.563	.440	.018	31	3,472
Exa Instant	.713	.409	.374	.273	31	12,020
Octen	.660	.063	.147	.157	31	12,020
DeepSeek	.608	.183	.184	.109	12	4,820
Ceramic	.011	.000	.000	.000	31	12,007

Table 6: Scholar recall@10 by query type, 31 runs, 29 September–28 October 2026. The last column sums all four types. Runs hold 332–420 queries each, 12,020 in total, and counts fall after account failures. Exa Lite has only one run. Bold marks the best engine per type.

all these paired differences exclude zero. They use common papers, unlike the available-case means in Table 6, and do not measure identifier-matcher error.

E Judge Sensitivity Checks

	κ					
Model	News	Rare	All	ρ	Exact	Mean
GPT-6 Luna	.88	.89	.89	.90	69	1.59
Claude Opus 5.5	.91	.90	.91	.93	71	1.67
Gemini 4 Argon	.90	.92	.91	.93	74	1.63
Grok 4.7	.91	.92	.92	.93	73	1.55

Table 7: Each model against the leave-one-out consensus of the other three on 1,194 results. κ is quadratic-weighted, ρ is Spearman, Exact is the share of identical grades in percent, and Mean is the mean grade.

Audit protocol. The 200 queries in Section 3.4 come from runs of 17–29 October with at least 19 engines, one day past the main window. Each model receives the production system prompt, the query profile, the URL, the title and the snippet at temperature 0 with high reasoning effort. Claude Opus 5.5 refuses the six results of one rare query, so all numbers use the other 1,194. κ and Exact use the consensus rounded to a grade, and ρ uses the mean. The 95% interval of GPT-6 Luna's κ from a query bootstrap is .87–.91, which overlaps the intervals of the other models. Pairwise κ is .86–.92. The new GPT-6 Luna grades agree with the stored production grades at .92, with 83% identical, so repeated runs of one judge already differ.

Literal quotes. Of 1,200 rare query-runs, 504 contain quotes. On unquoted queries, Exa, Perplexity and Brave (LLM ctx) lead, and Keenable is fourth. Replacing the rare score with its unquoted-only value gives overall $\tau_b = .96$ and a 3.2-point Keenable lead over Exa. Literal phrase matching therefore affects the rare ranking even when the overall order changes little. These checks use LLM judgments, not human labels.

F Finance Matching and Checks

Matching rules. A scale word fixes a money figure's scale. Bare figures, including parenthesized losses, can mean thousands or millions only when the preceding 200 characters contain a currency or financial cue. Years, percentages and multiples do not qualify. Diluted EPS also needs "EPS",

"per share" or "earnings per". Money gets no LLM backstop. Other fields can receive an upgrade, never a revoked hit. The full rules are in `src/needle/finance/canon.py`.

Why the narrow money band? An LLM audit of 399 results from production runs checks company, period and line item. Money-hit precision without table reading is .93 at the 0.1% band and .88 at 2%: the wider band admits more wrong periods and related line items. Money backstop upgrades have precision .36, which motivates excluding them. These are model judgments, not human gold labels.

Fresh sample. The fresh sample has 307 new queries (146 money, 161 other), tested on 17 engines. An LLM checks the first matching result for 335 table-reading hits, 100 exact-band money hits and 50 other-field hits, using at least the full 2,000-character input. Exact-band precision is .93, as in the production audit. Of 335 table-reading hits, 235 confirm the correct figure and period (.70), 95 omit period labels and five contradict the answer. Confirmation rates differ by engine: for example, 13/43 for Firecrawl and 42/44 for Exa. Pooled precision cannot establish each engine's error rate.

Effect on reported ranks. Discounting each engine's main-window table-reading hits by its unconfirmed fraction in the fresh sample lowers finance recall by up to 6.0 points. Dropping table reading entirely lowers finance recall by 0.1–7.5 points, with finance $\tau_b = .84$. In both checks, the overall order has $\tau_b = .98$ against Table 2 and overall ranks 1–10 stay fixed. The check bounds one scoring choice, not all matcher error. The fresh sample has more money queries than production and lacks the non-money backstop, so it cannot replace the main table.

G Prompts

The query-generation and judge prompts are Jinja templates in the public repository[1] at commit `2e48f194`, in the `prompts` folder of each task under `src/needle`. Scholar title queries use no prompt.

[1] https://github.com/keenableai/needle